

Pasindu Kalana Maduranga

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PROFESSIONAL SUMMARY

Senior AI Engineer & Tech Lead with 4+ years of experience architecting and delivering enterprise Generative AI (GenAI), Large Language Model (LLM) applications, autonomous Multi-Agent Systems, and standalone LLM observability platforms. Expert in multi-agent orchestration (LangGraph, Google ADK, CrewAI), Retrieval-Augmented Generation (RAG / GraphRAG), Natural Language Processing (NLP), and cost-governed multi-provider LLM infrastructure on LiteLLM proxy, achieving up to 60% operational efficiency improvements. Proven strategic problem solver serving as Acting Tech Lead and Consultant, leading cross-functional squads of 10+ engineers and mentoring 15+ developers, while instructing industry professionals and university students in Machine Learning (ML) and Agentic AI. Currently reading for an MSc in Cyber Security at the University of Wolverhampton, advancing research into AI security, prompt injection defenses, and penetration testing to build resilient, self-healing intelligent systems.

TECHNICAL SKILLS

- Generative AI, Agentic AI & LLMs:** Large Language Models (LLMs), Agentic AI, LangGraph, Google Agent Development Kit (ADK), CrewAI, Agent-to-Agent (A2A) Protocol, Retrieval-Augmented Generation (RAG), GraphRAG, Multi-Agent Orchestration, Prompt Engineering, Function Calling, Local LLM Fine-Tuning (LoRA / PEFT), Model Distillation
- Machine Learning & NLP:** Machine Learning (ML), Natural Language Processing (NLP), Deep Learning, Model Evaluation & Benchmarking, Hallucination Detection, Semantic Search, Synthetic Data Generation
- LLM Infrastructure, Governance & LLMOps:** LLMops, LiteLLM Proxy, Multi-Provider Model Routing (OpenAI GPT-4o, Anthropic Claude, Google Gemini/Vertex AI, AWS Bedrock, Azure OpenAI, Groq), Bring Your Own Key (BYOK) Management, Rate Limiting, Guardrail Enforcement, Token Cost Optimization, Vendor Abstraction Layers
- LLM Observability & Tracing:** Standalone LLM Tracking Engine, Framework-Agnostic Telemetry, Arize Phoenix, Tenant-Based Agentic Tracing, OpenTelemetry (OTel), Span Instrumentation, Latency Profiling
- Databases & Vector Search:** Vector Databases (Pinecone, Weaviate), Neo4j (Graph Databases, Knowledge Graphs), PostgreSQL, MongoDB, Vector Embeddings
- Backend & Distributed Systems:** Python (Advanced), Java (Spring Boot), Microservices Architecture, RESTful APIs, Event-Driven Architecture, Apache Kafka, Asynchronous Systems
- Cloud & DevOps / MLOps:** AWS (EC2, Lambda, S3, Bedrock), Docker, Kubernetes, Jenkins, CI/CD Pipelines, Harness CI/CD Migration, Linux
- Software Quality & Test Automation:** Selenium, JUnit, Automated Test Orchestration, Self-Healing Tests, Computer Vision / DOM Locators
- Cybersecurity & AI Safety:** AI Security, Penetration Testing, Metasploit, Nmap, Network Security, DevSecOps, Vulnerability Assessment, GDPR Compliance
- Leadership & Education:** Technical Leadership, Agile/Scrum, Squad Management, AI/ML & Agentic AI Instruction for Industry Professionals & University Students, Engineering Mentorship

PROFESSIONAL EXPERIENCE

Software Engineer (Acting Senior AI Engineer / Tech Lead) Jan 2023 – Present

Virtusa Pvt Ltd, Colombo, Sri Lanka

- Acting as Tech Lead and Consultant on enterprise Agentic AI and Generative AI (GenAI) initiatives — managing a squad of 10+ engineers, mentoring 15+ developers, conducting technical interviews, and driving executive buy-in with architectural blueprints, risk analyses, and ROI models.
- Engineered a **standalone LLM tracking** and observability engine compatible with any LLM architecture and orchestration framework (LangGraph, CrewAI, AutoGen, native APIs), delivering universal span instrumentation, prompt/response telemetry, token consumption metrics, and latency profiling.
- Served as squad Tech Lead and **strategic problem solver**, diagnosing and resolving complex framework constraints, multi-agent deadlocks, and latency bottlenecks across mission-critical systems.
- Architected an autonomous multi-agent test generation pipeline utilizing Neo4j knowledge graphs to parse documentation and map state transitions, generating test scenarios with synthetic data — reducing planning cycles by 60% with 90% edge-case coverage.
- Architected a multi-tenant LLM cost-governance and Bring Your Own Key (BYOK) platform on LiteLLM proxy: mapped tenant-project-user hierarchies with AES-256 encrypted credential storage across OpenAI, Anthropic, Google Gemini/Vertex AI, AWS Bedrock, Azure OpenAI, and Groq; enforced gateway prompt guardrails and per-team budgets.
- Contributed to an Agent-to-Agent (A2A) protocol-based agent platform — developing gateway routing, agent registry, and orchestrators for test-case, test-design, and test-script generation with authenticated, tenant-scoped routing and live reachability checks.
- Designed and deployed a full-stack observability platform on Arize Phoenix with custom OpenTelemetry span instrumentation, tenant-scoped trace isolation, and a token-spend dashboard for hallucination detection; boosted agent debugging efficiency by 55% and cut MTTR by 40%.
- Built a suite of LLM-powered test generators: an autonomous browser agent generating Selenium scripts with dynamic locators (70% faster development), a computer-vision/DOM-based self-healing agent for broken selectors (85% higher stability), and an OpenAPI/Swagger test generator.
- Built three GenAI failure-analysis tools: an ML+LLM hybrid classifier (65% faster root-cause analysis), a Google ADK triage system filtering flakes from bugs (80% less investigation time), and a Neo4j GraphRAG debugging assistant (50% faster fixes across 20+ teams).
- Delivered a context-aware synthetic test-data generator producing 10M+ GDPR-compliant records across 50+ projects, and fine-tuned/deployed local LLMs (LoRA/PEFT) achieving 92% accuracy for secure, offline code assistance without external API dependencies.
- Built an LLM-assisted CI/CD pipeline migration tool converting Jenkins, Azure DevOps, GitHub Actions, and GitLab pipelines into Harness YAML, integrating secret masking before LLM calls and automated validation/retry.

- Conducted technical training and hands-on workshops in AI, Machine Learning, and Agentic AI for corporate teams, industry professionals, and university students, translating advanced GenAI research into scalable engineering practice.

Software Engineer (Backend & Quality Engineering)

Dec 2021 – Dec 2022

Virtusa Pvt Ltd, Colombo, Sri Lanka

- Maintained 99.9% uptime for payment-processing microservices by engineering real-time Apache Kafka monitoring and automated Python alerting pipelines, reducing incident response time from 45 minutes to under 5 minutes.
- Refactored and expanded a 500+ test suite for a core enterprise application, establishing modular test patterns that enabled AI-driven test generation and averted a costly \$500K+ system rewrite.
- Developed custom Retrieval-Augmented Generation (RAG) tools automating routine code generation from internal technical documentation, lifting developer productivity by 35% and saving 10 hours per engineer weekly.
- Optimized AWS cloud infrastructure costs by 15% through intelligent query routing, directing simple requests to lightweight models while dynamically reserving frontier LLMs for high-complexity tasks.
- Streamlined cross-functional data workflows by 40% using automated Python and Groovy data-extraction scripts.

EDUCATION

MSc in Cyber Security

Present (Reading for MSc)

University of Wolverhampton

BSc (Hons) in Software Engineering — First Class Honours

2021 – 2022

Java Institute for Advanced Technology (Birmingham City University)

Professional Certificate in Java Technology

Awarded Dec 2023

Scottish Qualification Authority

Professional Higher Diploma in Software Engineering

2018 – 2019

Scottish Qualification Authority

Professional Diploma in Software Engineering

2017 – 2018

Scottish Qualification Authority

CERTIFICATIONS

- Certificate Programme on Cyber Security and Digital Forensics — SLIIT (Jan 2026)
- Virtusa Certified Agentic AI Engineer Program — Virtusa (Dec 2025)
- Oracle Certified Associate, Java SE 8 Programmer (Dec 2022)
- Professional Certificate in Java Technology (PCJT) — JIAT (Dec 2023)

TECHNICAL PROJECTS

- **Hierarchical Research Crew (CrewAI)** — Built a hierarchical multi-agent system to automate competitive market research with custom web-scraping tools. *Stack: Python, CrewAI, SerperDev, LLMs.*
- **Security Exploit Analysis Lab (Metasploit)** — Conducted penetration testing simulations to understand vulnerability exploitation and patch management. *Stack: Metasploit, Network Security, Penetration Testing.*
- **ShopMate (AI Shopping Assistant)** — Multi-agent e-commerce platform combining product search with Google ADK research agents. *Stack: Python, Google ADK, LiteLLM, Multi-Agent.*

AWARDS & ACHIEVEMENTS

- Innovator Award – Winner, Virtusa Delivery and Service Excellence Awards FY26 H2.
- Delivery Enabler Team Award – Winner, Virtusa Delivery and Service Excellence Awards FY26 H1.
- Delivery Enabler Team Award – Merit, Virtusa Delivery and Service Excellence Awards FY26 H2.
- 1st Place — SRE Hackathon, Virtusa Pvt Ltd (2021) — Awarded for excellence in Site Reliability Engineering.
- Conference Speaker — Presented “Building Self-Healing Systems with Agentic AI” at internal tech summit (2024).
- AI Educator & Technical Mentor — Taught AI, ML, and Agentic AI to industry professionals and university students; mentored 15+ junior developers on Agentic AI patterns, LangGraph, and RAG implementation.
- Participant — Virtusa Agentic AI Engineer Hackathon.